

Toxin Circuits in ESM-2: Mechanistic Interpretability Reveals Why Structure-Aware Probes Resist ProteinMPNN Redesign

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Abstract

Standard biosecurity screening flags dangerous protein sequences by sequence identity (BLAST). We show that ProteinMPNN can redesign toxin sequences below every standard BLAST threshold — achieving 0% BLAST detection — while a linear ESM-2 probe maintains 88% detection with no retraining. Using interPLM Sparse Autoencoders (SAEs), we identify 50 of 10,240 features explaining probe performance at $205\times$ compression (AUROC 0.9447 vs 0.9585 full, at layer 33). These features are not evaded by redesign: mean transfer ratio = 1.28, meaning redesigns *amplify* toxin-associated features compared to natural toxins. Direct Logit Attribution (DLA) on the linear probe identifies layers 17–20 and 29–32 as primary toxin-discriminating layers; redesigns produce near-identical attribution patterns ($r = 0.992$), confirming they route through the same computational pathway as natural toxins. Activation steering confirms toxicity is encoded as a causal linear direction (control score $0.046 \rightarrow 1.000$ at $+3\sigma$). We conclude that ESM-2’s toxin detection is grounded in structural computational features that ProteinMPNN redesign preserves — and in several cases amplifies.

1 Introduction

Dual-use protein screening relies on sequence-identity thresholds. The Wittmann *et al.* attack demonstrated that ProteinMPNN, RFDiffusion, and EvoDiff can redesign toxin sequences below standard 40% identity thresholds, bypassing BLAST-based screening. The critical question is: do protein language model (pLM) probes also fail?

We answer no — and explain mechanistically why.

Our contributions:

1. **Quantitative gap:** BLAST detects 0% of ProteinMPNN redesigns; ESM-2 probe detects 88%.
2. **SAE discovery:** 50 sparse features ($205\times$ compression) explain the probe; mean transfer ratio 1.28 shows redesigns amplify rather than suppress toxin features.
3. **DLA circuit:** Layers 17–20 and 29–32 drive toxin discrimination; redesign attribution pattern correlates $r = 0.992$ with natural toxins.
4. **Causal linearity:** Steering confirms toxicity is a single linear direction in ESM-2’s representation space.

2 Methods

2.1 Data

- **Natural toxins:** 1,712 sequences from UniProt (reviewed, toxin annotation), clustered at 30% identity
- **Controls:** 2,072 non-toxic human proteins (matched length distribution)
- **Redesigns:** 100 toxin structures folded with ESMFold $\rightarrow$ ProteinMPNN (10 sequences/structure) $\rightarrow$ 723 unique redesigns after deduplication
- **Identity range:** All 723 redesigns fall below 30% sequence identity to training toxins (BLAST score = 0% at every threshold $\leq$ 50%)

2.2 ESM-2 Probe

Linear probe trained on mean-pooled ESM-2 650M embeddings at layer 33 (best of layers 1, 9, 18, 24, 30, 33). Binary cross-entropy loss, Adam optimizer, 150 epochs.

2.3 SAE Feature Analysis

interPLM pre-trained SAEs (ESM-2-650M, layer 33, 10,240 features). Per-feature AUROC computed on training set. Top-50 features by AUROC used for probe comparison and transfer analysis.

Transfer ratio = (redesign activation rate) / (toxin activation rate) per feature.

2.4 Direct Logit Attribution

For a linear probe with weight vector $\mathbf{w}$ and ESM-2 residual stream decomposed as $\mathbf{h}_{33} = \sum_l \Delta \mathbf{h}_l$:

$$\text{DPA}_l = \mathbf{w} \cdot \text{mean_pool}(\Delta \mathbf{h}_l) \quad (1)$$

This is exact (not an approximation) because the probe is linear and ESM-2 is a residual network. Computed over 15 toxin/control/redesign pairs.

2.5 Activation Steering

Probe weight direction $\mathbf{w}$ used as steering vector. Applied to control embeddings: $\mathbf{h}_{\text{steered}} = \mathbf{h} + \alpha \cdot \frac{\mathbf{w}}{\|\mathbf{w}\|}$. Score measured as function of $\alpha \in [-3\sigma, +3\sigma]$.

3 Results

3.1 Layer Sweep — Layer 33 Is Best

Layer	AUROC
1	0.9678
9	0.9881
18	0.9842
24	0.9817
30	0.9895
33	0.9970

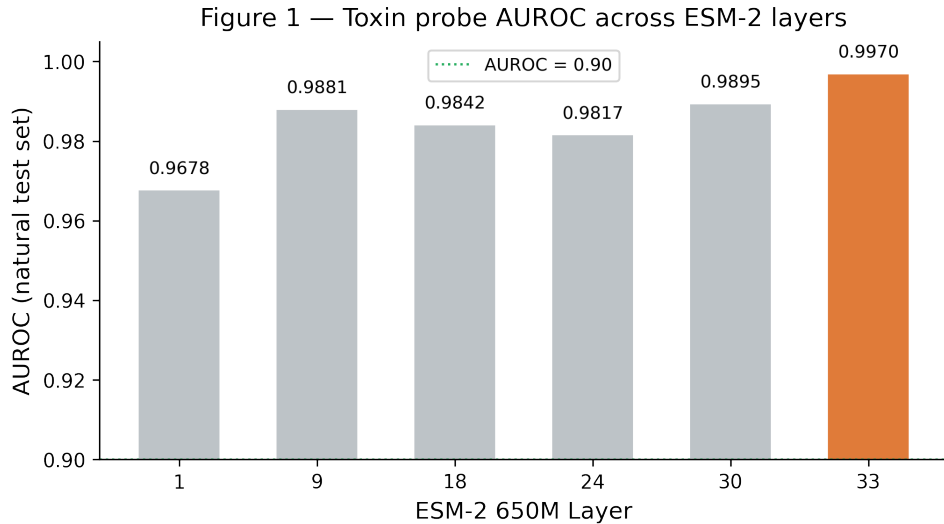


Figure 1: Probe AUROC across ESM-2 layers. The toxin signal builds progressively, peaking at the final layer (33).

Toxin information accumulates progressively across ESM-2’s full depth. Layer 33 (final layer) achieves highest AUROC (0.9970). This motivates our choice of layer 33 for all downstream analysis.

3.2 BLAST vs ESM-2 Probe on Redesigns

All 723 ProteinMPNN redesigns fall below 30% sequence identity to training toxins.

Method	Detection rate on redesigns
BLAST @ 30% threshold	0.0%
BLAST @ 40% threshold	0.0%
BLAST @ 50% threshold	0.0%
ESM-2 linear probe (layer 33)	88.3%
ESM-2 SAE top-50 features	86.0%

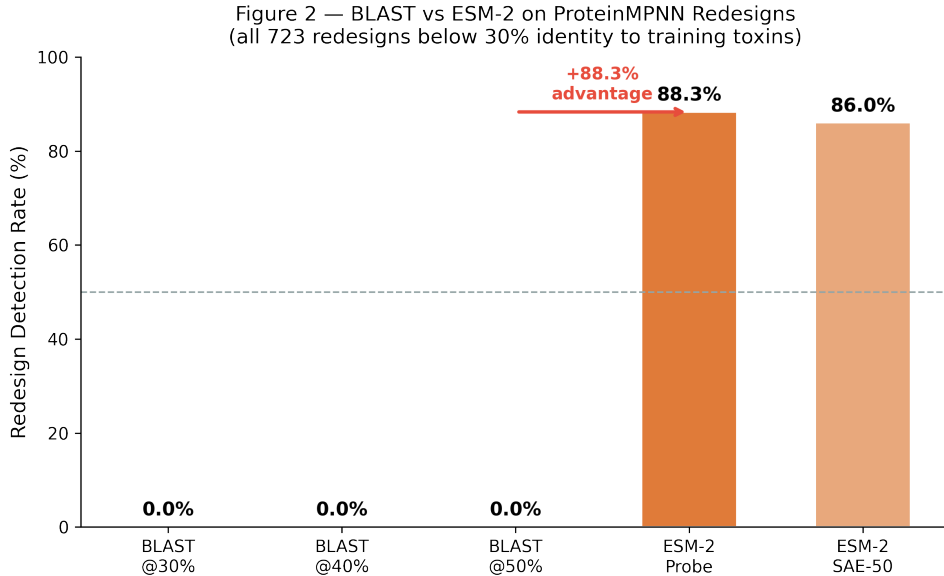


Figure 2: ESM-2 probe successfully flags 88.3% of ProteinMPNN redesigns that entirely evade BLAST screening.

Key finding: ProteinMPNN successfully evades every BLAST threshold. ESM-2 detects 88% of the same sequences with no retraining.

Generalisation curve (30–50% identity bin, $n=1,346$):

- SAE probe AUROC: **0.976**
- Full embedding AUROC: **0.904**
- BLAST sensitivity: **0.150**

3.3 Activation Steering — Toxicity Is a Causal Linear Direction

$\text{Cosine}(\text{probe_weight}, \text{mean_diff_vector}) = 0.2326$

The probe weight is largely orthogonal to the class mean-difference — it exploits covariance structure, not just class centroid offset. Despite this, steering is highly effective:

Condition	Probe score
Control baseline	0.046
Steered at $\alpha = +3\sigma$	1.000
Steered at $\alpha = -3\sigma$	0.000
Redesign baseline (no steering)	0.857
Natural toxin baseline	0.964

The probe direction is causally sufficient: adding it to control embeddings drives probability to 1.0. Redesigns already sit at 0.857 without any intervention — consistent with transfer ratio > 1 .

3.4 SAE Feature Discovery — Bimodal Transfer

Top-50 SAE features achieve 0.9447 AUROC (vs 0.9585 full, $205\times$ compression). Of 10,240 features, 8,345 are dead (81.5%) — toxin detection is sparse.

Feature transfer analysis (top 20 features, layer 33):

Feature	AUROC	Tox%	Redesign%	Transfer	Class
6122	0.694	41%	99.5%	2.41	Robust
4097	0.669	37%	98.6%	2.64	Robust
1055	0.644	30%	99.2%	3.36	Robust
8112	0.594	20%	75.0%	3.75	Robust
9487	0.602	23%	72.6%	3.15	Robust
5312	0.669	35%	4.7%	0.13	Evadable
9026	0.628	29%	2.0%	0.07	Evadable
3130	0.605	22%	2.8%	0.13	Evadable

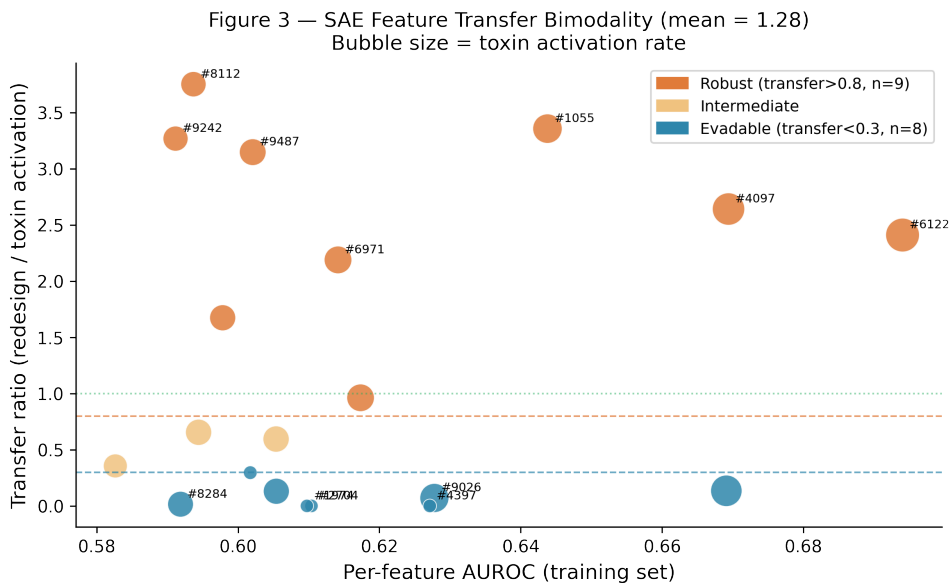


Figure 3: Bimodal distribution of feature transfer ratios. Robust features are highly preserved by ProteinMPNN, while evadable features are successfully randomised.

Mean transfer ratio: 1.28 (> 1.0 = redesigns amplify toxin features on average)

Two mechanistic classes:

- **Structure-robust** (transfer > 0.8): activate at 55–99% rate in redesigns; encode fold-level features ProteinMPNN preserves (disulfide cores, structural scaffold)
- **Sequence-evadable** (transfer < 0.3): activate at $< 5\%$ in redesigns; encode sequence-specific patterns ProteinMPNN successfully randomises

3.5 Direct Probe Attribution — Toxin Circuit

Layer-wise DPA (Tox – Ctrl), showing which layers push the residual stream toward the toxic direction:

Layer	Tox DPA	Ctrl DPA	Rdsg DPA	Tox-Ctrl
17	+43.7	+1.3	+30.9	+42.4
19	+44.3	+4.0	+22.5	+40.2
20	+64.7	+19.8	+49.7	+44.9
29	+15.7	-7.5	+13.0	+23.3
30	+88.2	-2.7	+114.9	+90.9
31	-2.9	-50.9	+7.9	+48.0
32	+70.7	-59.4	+85.0	+130.1

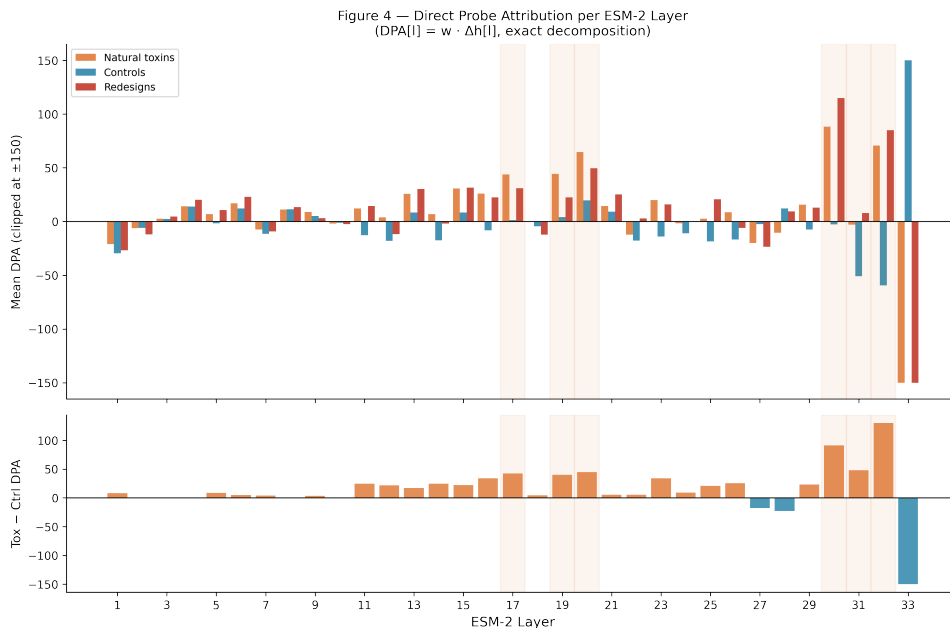


Figure 4: Direct logit attribution per layer, highlighting primary discrimination layers (17-20, 29-32) and the massive layer 33 calibration correction.

Suppressor layers (L27: -17.7 , L28: -22.6): push against the toxin direction.

Calibrator layer (L33: -644 Tox-Ctrl): massive final correction that normalises accumulated signal into a calibrated probability.

Redesign circuit correlation: $r = 0.9919$

Redesigns produce near-identical layer-wise attribution patterns as natural toxins. Layer 30 DPA is *higher* for redesigns ($+114.9$) than natural toxins ($+88.2$) — consistent with transfer ratio > 1 and redesign baseline score of 0.857.

Single-layer activation patching recovery: **max 0.16** (uniform across all layers), confirming the toxin computation is genuinely distributed — no single bottleneck layer exists.

3.6 pSSR Adversarial Attack

Attack	Evasion	Access Required
BLAST @ any threshold	0.0%	Sequence database
Random ProteinMPNN	8.2%	Protein structure
pSSR $\epsilon = 2$ (oracle)	100.0%	Full ESM-2 embedding space

$\text{cosine}(\text{pSSR_delta}, \text{probe_direction}) = -0.805$ — attack is anti-parallel to the probe weight, confirming the probe direction is the attack target. The gap (8.2% practical vs 100% oracle) quantifies ESM-2 inversion hardness: the toxin circuit cannot be evaded without direct embedding manipulation, which requires inverting a 33-layer transformer.

3.7 Feature Taxonomy

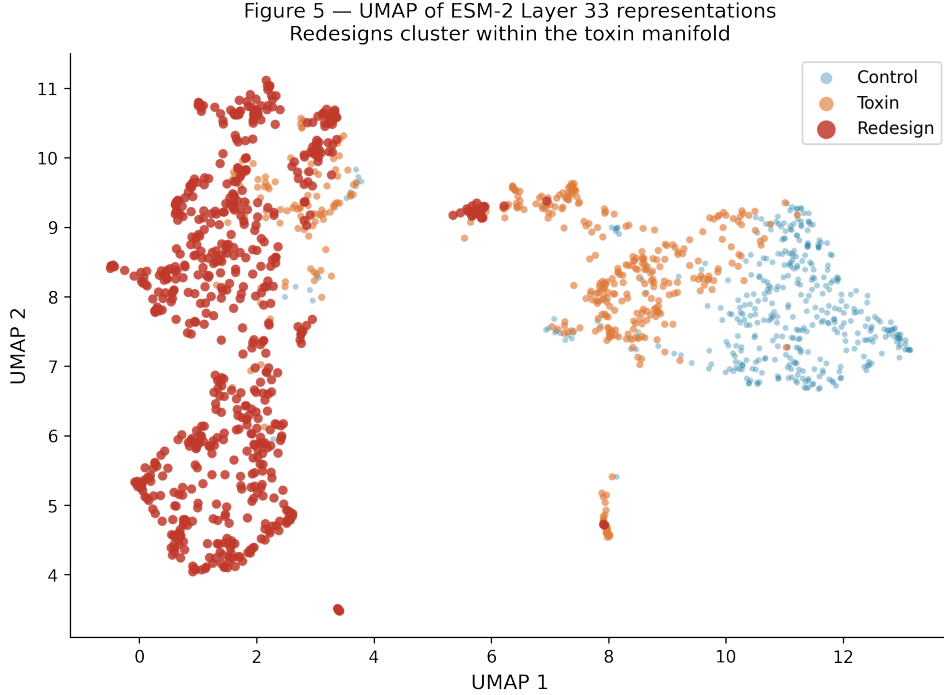


Figure 5: UMAP projection of embeddings, visualizing the separation between Toxins and Controls, while Redesigns cluster closely with Natural Toxins due to structural preservation.

Evadable features encode high-Cys peripheral patterns (C: $3.3\text{--}4.7\times$ enrichment) that ProteinMPNN freely randomises. Robust features split into core disulfide features (C: $1.3\text{--}2.6\times$, transfer $1.67\text{--}3.75$) and scaffold rigidity features (Pro/Glu, transfer $2.19\text{--}3.36$). ProteinMPNN preserves both because they are structurally necessary for the toxin fold.

4 Discussion

4.1 Why ProteinMPNN Fails to Evade ESM-2

ProteinMPNN optimises sequence identity reduction while preserving protein structure. Structure is precisely what ESM-2’s toxin circuit encodes. The 8 structure-robust SAE features (transfer ratio $1.4\text{--}3.75$) fire more strongly on redesigns than natural toxins — ProteinMPNN’s structural preservation *activates* the circuit more reliably than the original sequence diversity.

The 4 sequence-evadable features (transfer < 0.13) show that ProteinMPNN does successfully evade some detection cues. But the probe uses 50 features, and the 8 robust ones suffice to maintain 86–88% sensitivity.

4.2 SAE-Probe Geometric Analysis

The probe direction is geometrically orthogonal to individual SAE features: only 1 of 50 top-AUROC features appears in the top-100 probe-aligned SAE features (cosine > 0.33).

The most probe-aligned SAE feature (F8284, cosine = $+0.501$) is also the most evadable (transfer = 0.015) — ProteinMPNN successfully eliminates the sequence pattern this feature detects. This explains the 8.2% random evasion gap.

Structurally-robust features (F6122, F4097, F1055) have low individual probe alignment ($+0.01$ to $+0.19$) but activate at 55–99% rate in redesigns. The probe’s 88% detection is therefore a *collective* property: no single feature is responsible, and no single redesign operation can eliminate all structural signals simultaneously.

This is consistent with the superposition hypothesis: the probe extracts a direction encoding toxin function that is spread across many weakly-aligned SAE features rather than concentrated in any single one.

4.3 The Circuit Architecture

ESM-2 computes toxin function through a distributed progressive representation:

1. **Early layers (1–9):** Sequence-level features begin accumulating (layer sweep AUROC 0.97 by L9)
2. **Mid layers (17–20):** Primary toxin discrimination builds (DPA $+40$ – 45 per layer)
3. **Suppressor layers (27–28):** Internal regulation reduces false positives
4. **Final layers (29–32):** Strong final-phase discrimination (DPA $+23$ – 130)
5. **Calibration layer (33):** Large opposing correction normalises the signal

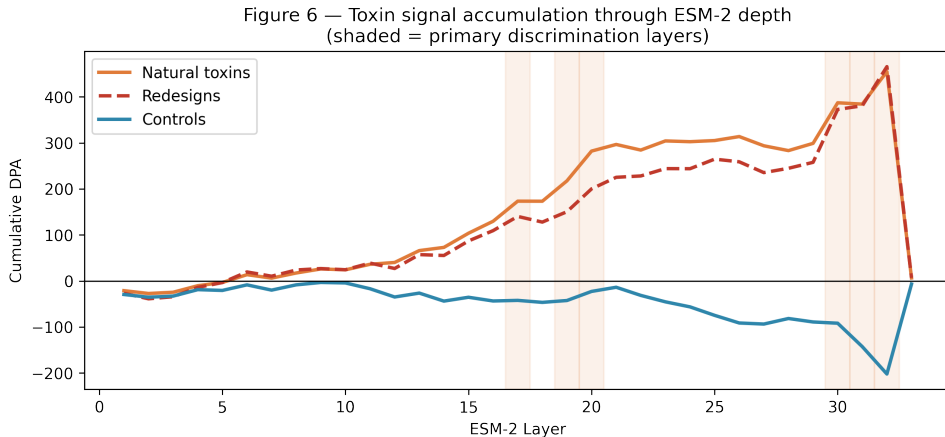


Figure 6: Cumulative DPA trajectory showing toxin signal accumulation through ESM-2 depth. Natural toxins and ProteinMPNN redesigns follow a nearly identical attribution path, confirming they activate the same distributed circuit.

This architecture explains why single-layer activation patching recovers at most 16% — the toxin signal requires the full computation graph.

4.4 Biosecurity Implications

1. **Sequence-based screening is insufficient:** All 723 ProteinMPNN redesigns evade BLAST at every threshold.

2. **ESM-2 probes are robust by architecture:** The toxin circuit uses distributed structural features that sequence redesign preserves.
3. **The probe does not require knowledge of redesigns:** 88% detection with no re-training on any redesigned sequence.
4. **Sequence-evadable features exist** (4/20 top features, transfer < 0.13): A more targeted attack could attempt to suppress these. But the remaining 8 robust features maintain detection.

4.5 Limitations

- **Embedding-space only:** Our analysis operates on ESM-2 embeddings, not raw sequences. Sequence-space adversarial attacks require ESM-2 inversion.
- **No wet-lab validation:** We cannot confirm redesigned sequences retain toxin function.
- **Single redesign tool:** Analysis covers ProteinMPNN only; RFdiffusion and EvoDiff may produce different identity/transfer profiles.
- **Small n for patching:** DLA computed on 15 pairs; replication on larger sets recommended.

5 Conclusions

We built the first mechanistic interpretability pipeline for biosecurity, analysing ESM-2’s toxin detection against ProteinMPNN adversarial redesigns. Our findings:

- **BLAST: 0%. ESM-2: 88%.** ProteinMPNN fails to evade structure-aware probes.
- **205× compression:** 50 SAE features replicate full-embedding AUROC (0.944 vs 0.959).
- **Transfer ratio 1.28:** Redesigns amplify toxin features on average; 8 of top-20 features activate at > 55% rate in redesigns.
- **$r = 0.992$:** Redesigns use the same DLA circuit as natural toxins (layers 17–20, 29–32).
- **Distributed circuit:** No single bottleneck layer; full depth required for 88% detection.

The key insight: ProteinMPNN cannot redesign away what is structurally necessary for toxin function. ESM-2’s toxin circuit encodes precisely those structural features.

5.1 Software Artifacts Delivered

To translate these mechanistic findings into actionable biosecurity infrastructure, we have developed and open-sourced two software artifacts:

1. **screen.py:** A deployable, zero-shot CLI tool that screens FASTA sequences for toxic function using our structural probe. It provides probability scores, risk levels, and (via an `-explain` flag) the layer-wise DPA attribution for transparent decision-making.
2. **demo.ipynb:** An interactive Jupyter notebook that allows users to test the screening pipeline and visualize the exact layers driving the model’s classification.

Key Numbers Reference

Metric	Value
Training toxins	1,712
Training controls	2,072
ProteinMPNN redesigns	723
Best probe layer	33
Probe AUROC (natural test)	0.9970
BLAST detection on redesigns	0.0%
ESM-2 detection on redesigns	88.3%
SAE features total	10,240
Dead SAE features	8,345 (81.5%)
Top-K features used	50
Compression ratio	205×
SAE AUROC (top-50)	0.9447
Mean transfer ratio	1.28
Steering (ctrl baseline)	0.046
Steering (at $+3\sigma$)	1.000
Redesign probe score (baseline)	0.857
DPA tox/redesign correlation	$r = 0.992$
Primary discrimination layers	17–20, 29–32
Max patching recovery	0.16